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Department of Computational & Data Sciences
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DREAM:Lab

Quantum-Classical Workflow Patterns

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INDIAN INSTITUTE OF SCIENCE
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⟨SPAARQ⟩
Software Platform and Algorithms to Accelerate Research in Quantum-HPC

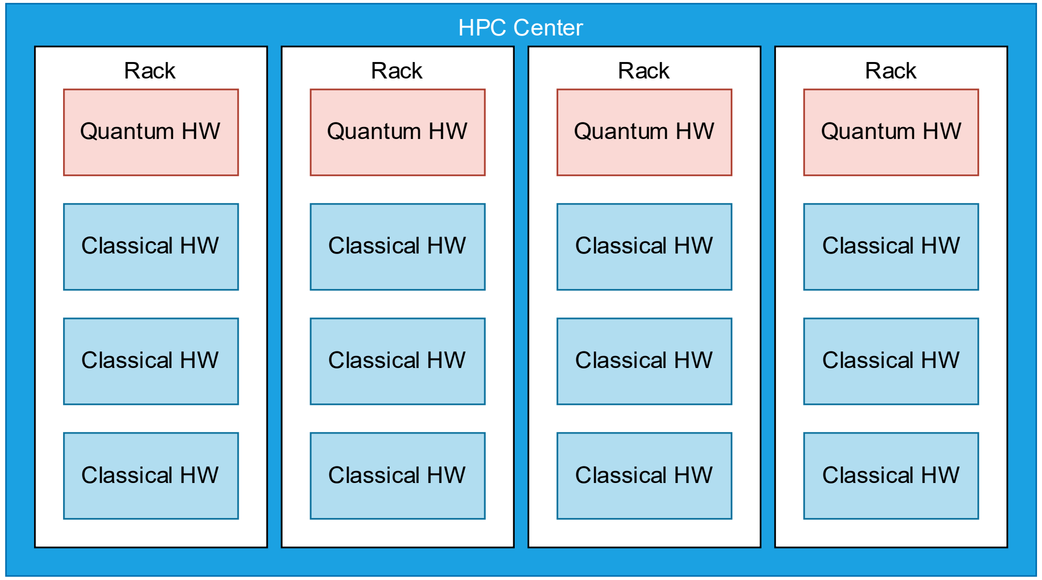
Workflow Patterns & Integration Patterns

- **Workflow Patterns** describe a **recurring structure** in the flow of **data and control** of a hybrid quantum-classical (HQC) application.
  - Characterize the kind of data that flows between classical and quantum components, the sequence of operations performed on it.

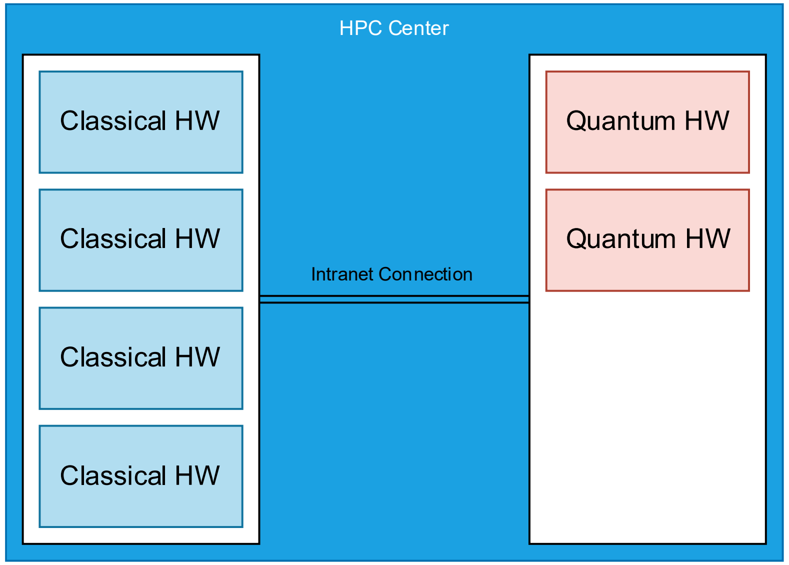
- **Integration Patterns** describe **how workflow patterns are mapped** to classical and quantum infrastructure, how components are **deployed and coupled**.
  - Characterize the latency, bandwidth, communication channels and synchronization between different computing components.

Deployment Models

On-Prem (Near Quantum)



Co-located, tight integration

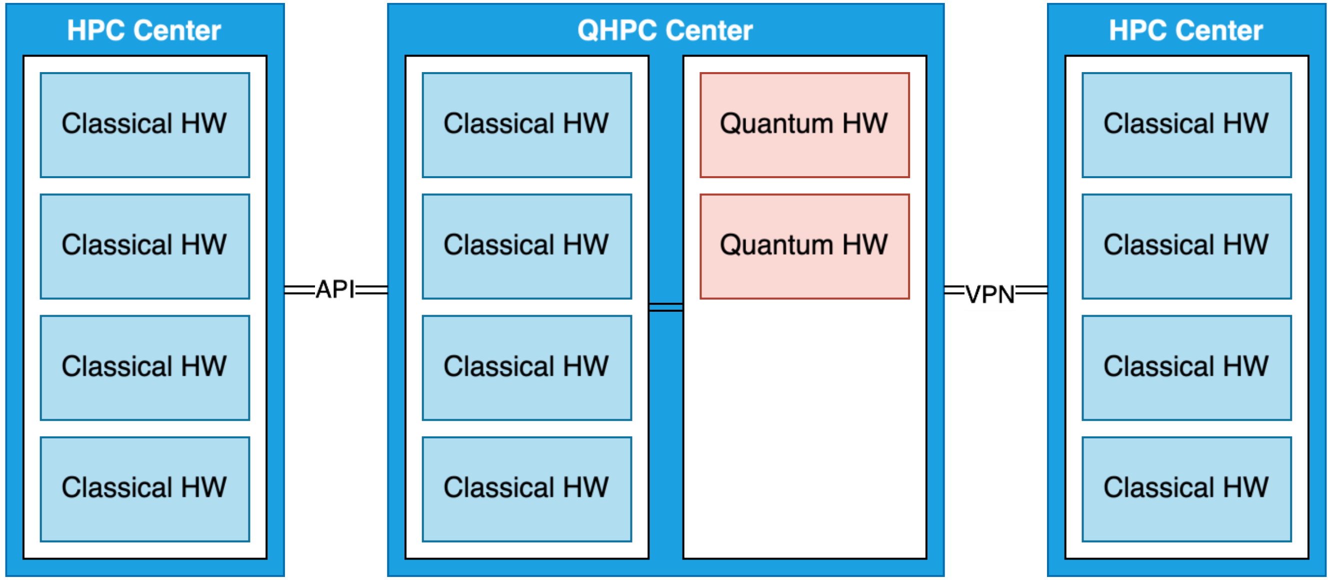


Co-located, loose integration

K. A. Britt and T. S. Humble, "High-Performance Computing with Quantum Processing Units," ACM Journal on Emerging Technologies in Computing Systems, vol. 13, no. 3, pp. 1–13, Mar. 2017, doi: 10.1145/3007651.

Deployment Models

Off-Prem (Far Quantum)

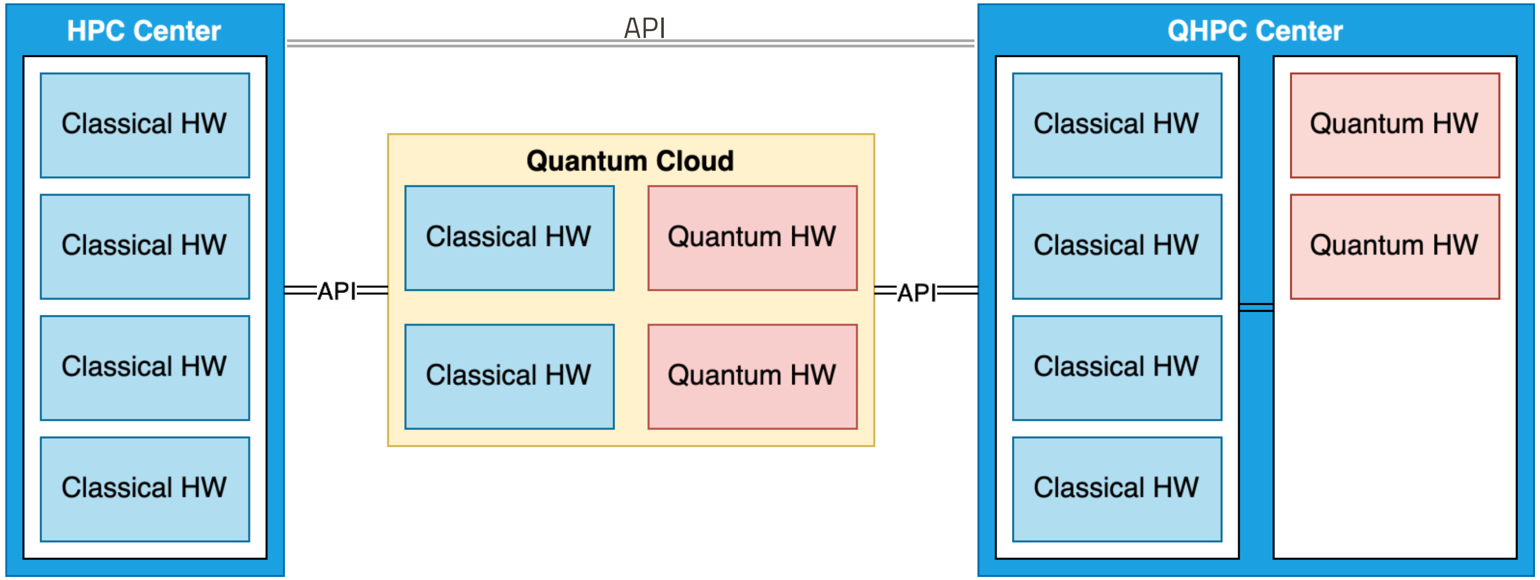


Far-Quantum integration model

K. A. Britt and T. S. Humble, "High-Performance Computing with Quantum Processing Units," ACM Journal on Emerging Technologies in Computing Systems, vol. 13, no. 3, pp. 1–13, Mar. 2017, doi: 10.1145/3007651.

Deployment Models

Off-Prem (Far Quantum)



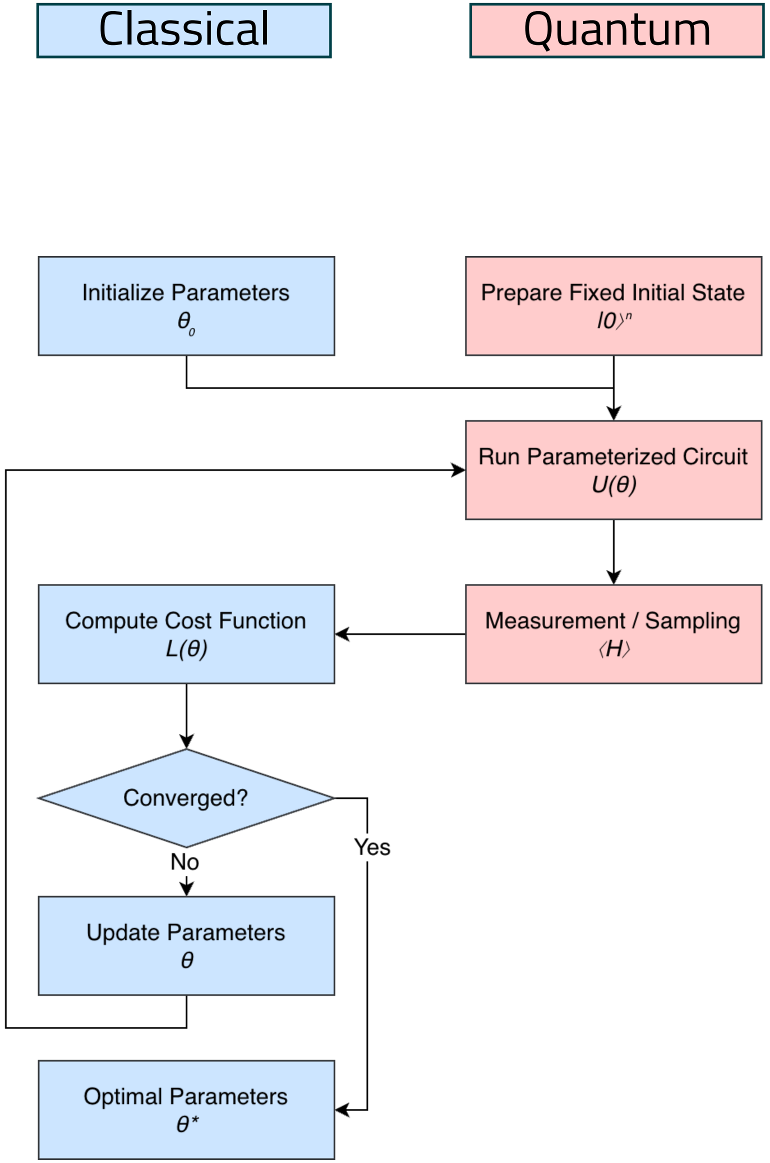
Far-Quantum integration model (including cloud)

K. A. Britt and T. S. Humble, "High-Performance Computing with Quantum Processing Units," ACM Journal on Emerging Technologies in Computing Systems, vol. 13, no. 3, pp. 1–13, Mar. 2017, doi: 10.1145/3007651.

Workflow Patterns

Variational Iterative Optimization

- Iteratively optimizes parameters to minimize a cost function
- QPU executes *fixed, parameterized circuit* repeatedly
- Classical optimizer *updates parameters* based on measurements and loss
- Requires complete classical–quantum round-trip for each iteration

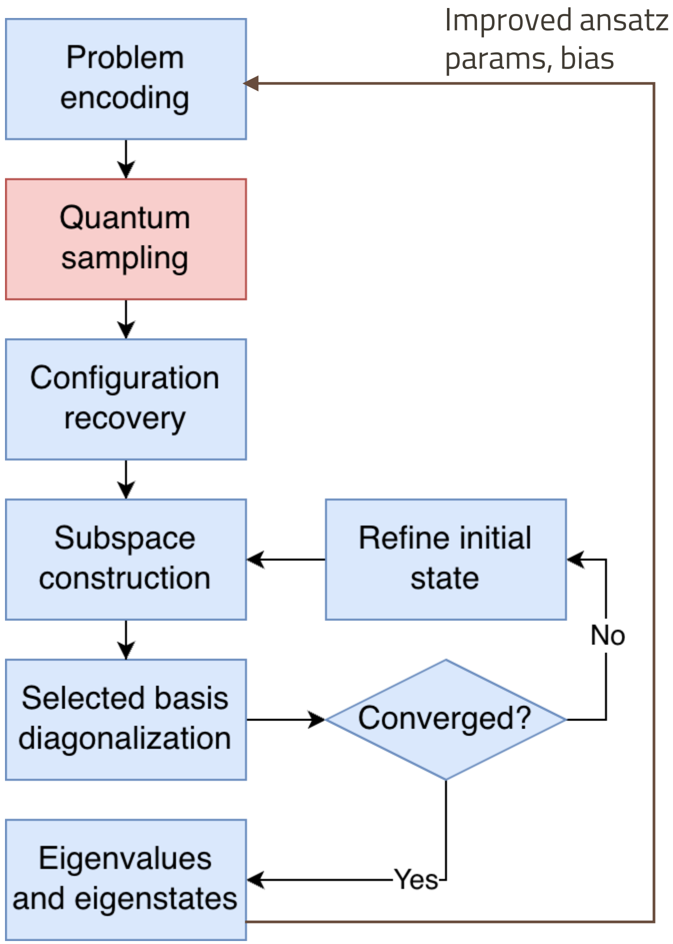


"Variational circuits — PennyLane." https://pennylane.ai/qml/glossary/variational_circuit (accessed Apr. 27, 2026)

Workflow Patterns

Sample-based Quantum Diagonalization (SQD)

- Sampling-driven workflow
- Reconstructs eigenvalues and eigenstates of a Hamiltonian from quantum measurement samples
- QPU produces measurement *samples* from a fixed or weakly parameterized circuit
- Classically *solves* a reduced-space eigenvalue problem *(compute intensive, HPC requirement)*
- *Optional iteration to improve sampling quality*

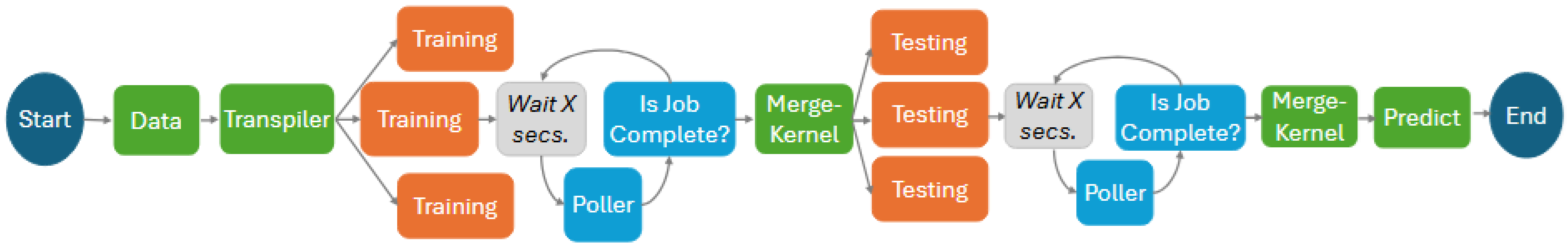


"Sample-based Quantum Diagonalization | IBM Quantum Learning," IBM Quantum Learning. https://www.ibm.com/quantum/blog/riken-fugaku-qcsc, https://arxiv.org/html/2511.00224v1

Workflow Patterns

Hybrid QSVM Training & Prediction

- Classical data prep: Build train/test splits + feature/label encoding
- Classical circuit prep: Generate parameterized <kernel circuits, shard> pairs
- Hybrid training-kernel stage: Per-shard classical transpile → quantum execution → emit kernel shards (parallel fan-out)
- Classical learning/merge: Merge shards into full kernel, then train classical SVM
- Hybrid testing/inference: Per-shard classical transpile → quantum execution, then merge test-kernel shards and predict/score
- Submit jobs + async poll/wait (classical orchestration around queued quantum runs)

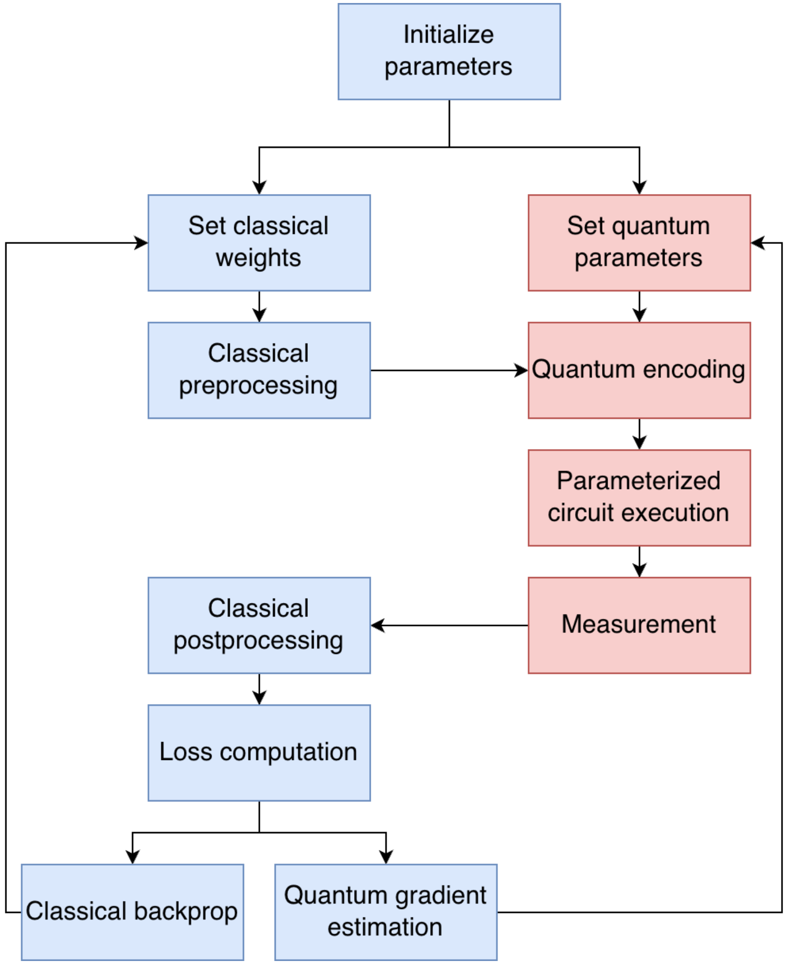


Jha, Srivastava, Majumdar, Seshadri, Ray and Simmhan, Orchestration and Profiling of Hybrid Quantum–Classical FaaS Workflows on Multi-Cloud Platforms, IEEE CCGRID Extended Version, 2026

Workflow Patterns

Hybrid QNN Training

- Trains parameterized quantum circuits in combination with classical layers
- QPUs execute encoded circuits and produces expectation values as outputs
- Classical components compute cost functions based on quantum measurements
- No direct quantum state persistence across iterations
- Barren plateaus and encoding pose challenges in implementation

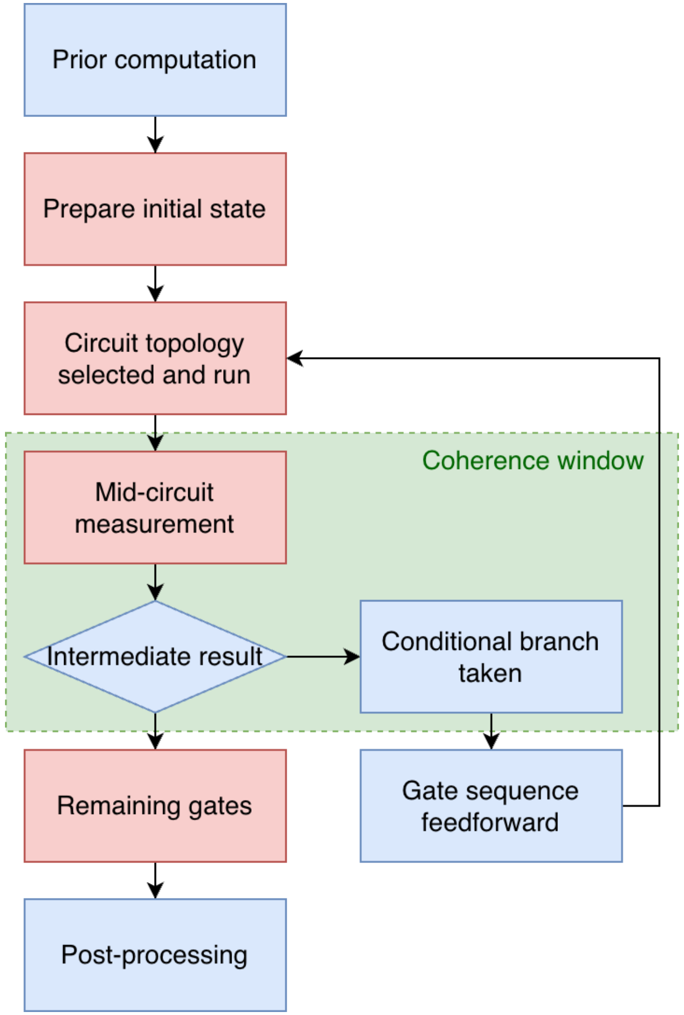


G. Verdon et al., "Learning to learn with quantum neural networks via classical neural networks," arXiv (Cornell University), Jul. 2019, doi: 10.48550/arxiv.1907.05415.

Workflow Patterns

Dynamic Circuit Generation

- Circuit topology is decided during runtime
- QPU runs a circuit that changes based on mid-circuit measurements or conditional gates
- Classical control mechanisms must run within the coherence time of qubits
- Intra-circuit feedback *informs the later structure of the circuit*

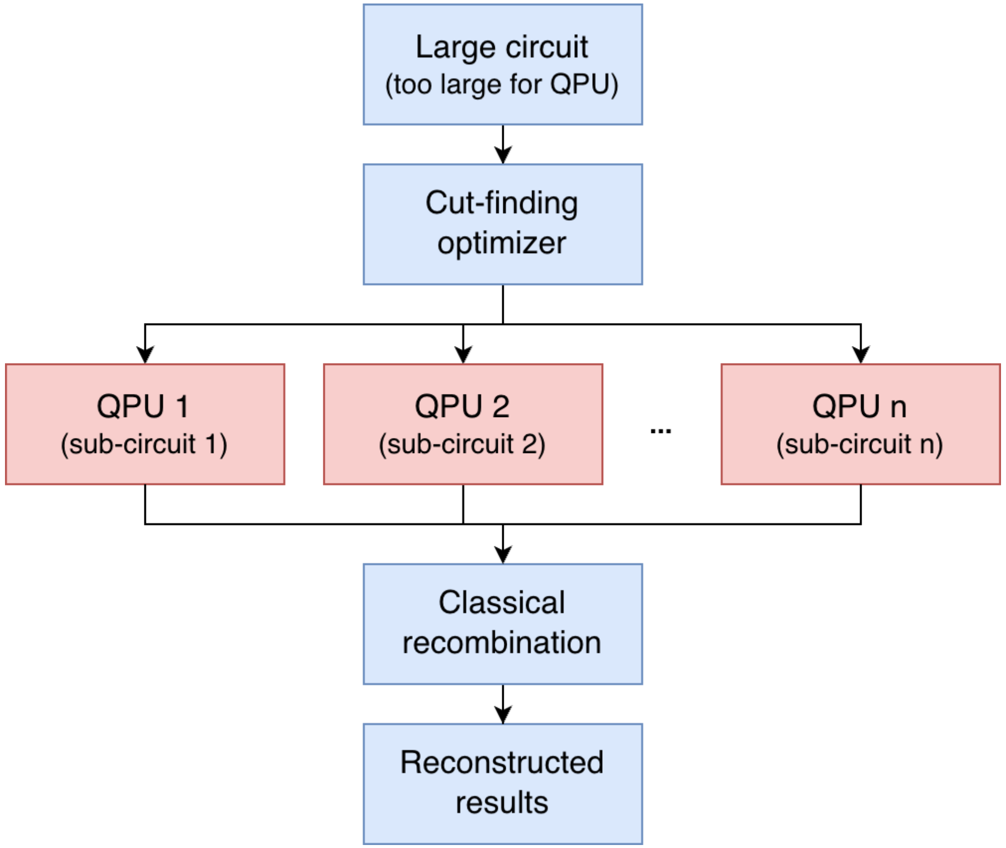


H. Buhrman, M. Folkertsma, B. Loff, and N. M. P. Neumann, "State preparation by shallow circuits using feed forward," arXiv (Cornell University), Jul. 2023, doi: 10.48550/arxiv.2307.14840.

Workflow Patterns

Circuit Cutting & Knitting

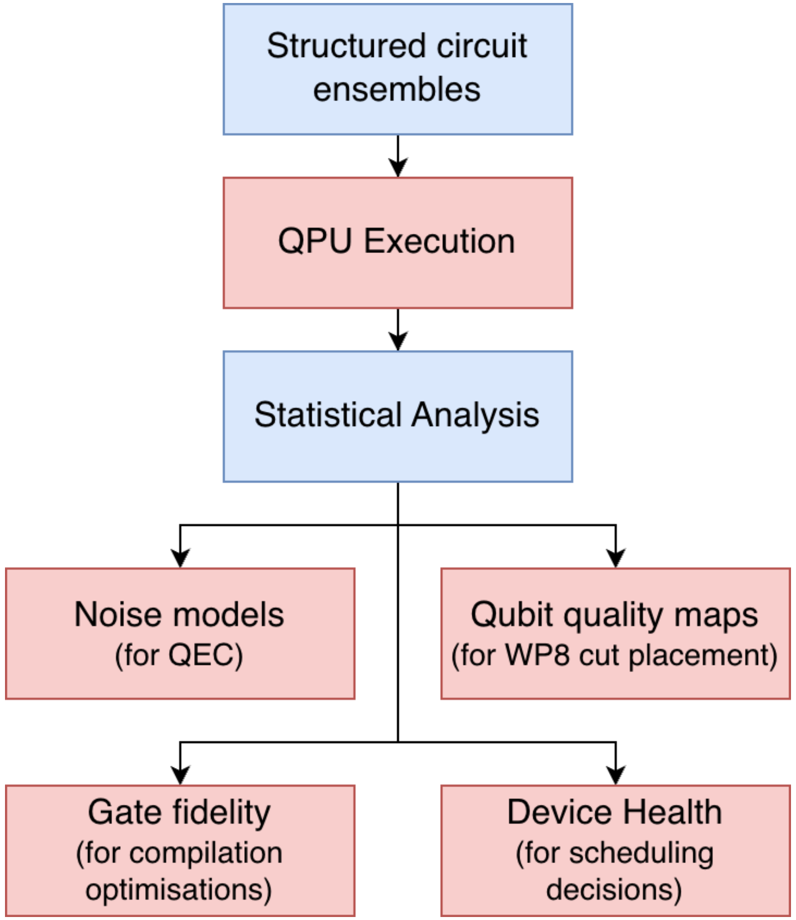
- Dynamically partitions a large circuit into smaller subcircuits
  - Splits a large circuit into smaller fragments that fit qubit limits, can be executed concurrently, reduce end-to-end latency
- Classical optimizer decides & makes the cuts
- Fan-out execution of sub-ciruits on (different) QPUs
- Classical post-processor combines sub-results using quasi-probabilistic decomposition
- Optimal cut-finding remains challenging due to exponential overhead



Jha, Srivastava, Pal, Manoj, Majumdar, Khare, Seshadri, Kulkarni, Ray and Simmhan, Choreography and Profiling of Quantum-Classical FaaS Workflows On Hybrid Clouds, *IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGRID)*, 2025

Device Characterization Circuits

- Ensembles of benchmarking circuits run on QPUs
  - **Quality Indicator Circuits (QIC):** Lightweight probe circuit synthesized from user circuit, whose ideal noiseless outcome is known
- Classical analysis models fidelity, error rates, noise models, etc.
  - Used to decide layout of user circuit
- Device calibration data drifts and needs to be *refreshed periodically*

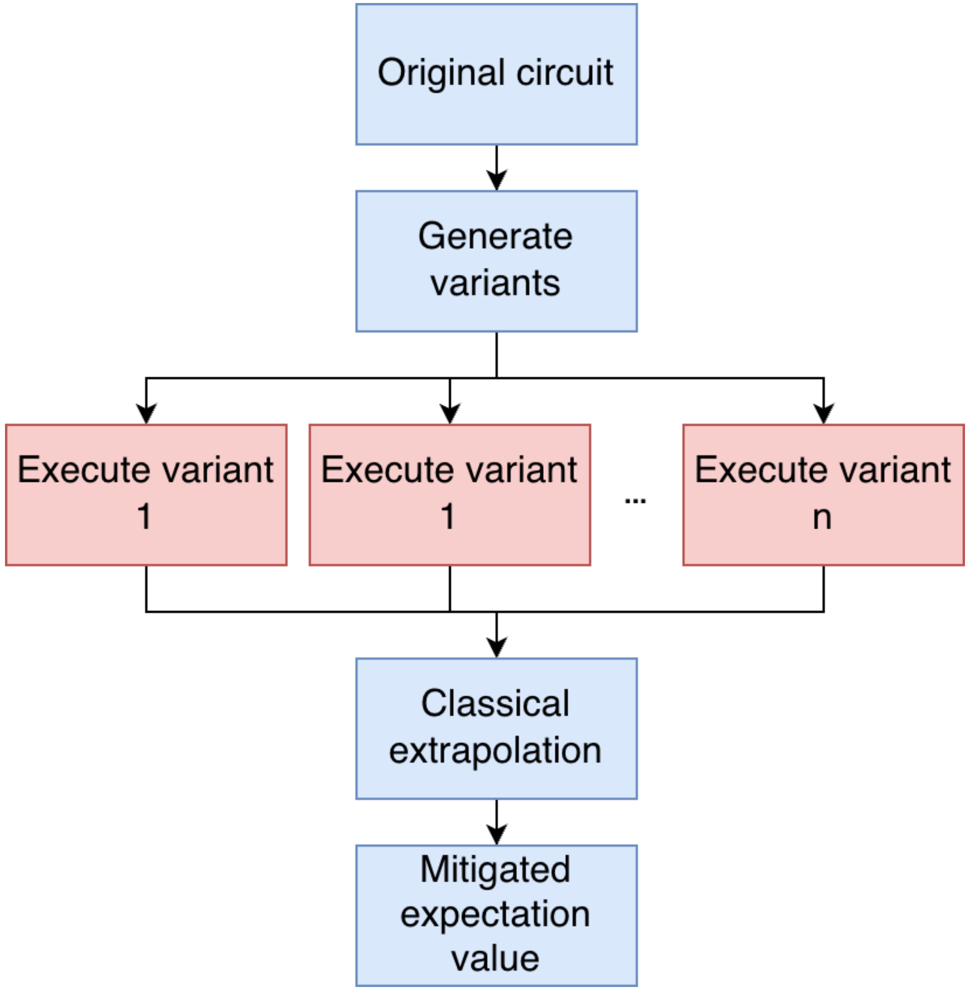


S. Srivastava, R. Majumdar, P. V. Seshadri, A. Ray, and Y. Simmhan, "Lightweight Targeted Estimation of Layout Noise in a Quantum Computer using Quality Indicator Circuits," arXiv, Sep. 23, 2025. https://arxiv.org/abs/2509.18679

Workflow Patterns

Error Mitigation Ensembles

- Wraps over target workflows to mitigate errors
- QPUs execute variants of a circuit at different noise levels, with different twirling instances, or in different measurement bases
- Results are classically aggregated
- Applies techniques such as ZNE, PEC, over target workflows to mitigate noise



D. Pal and Y. Simmhan, "Folding-Free Zero-Noise extrapolation by layout-induced noise diversity," arXiv, Mar. 14, 2026. https://arxiv.org/abs/2603.13949

More being accumulated & characterized…

- Markov Chain Monte Carlo (IBM), https://arxiv.org/abs/2602.06171v2
- 6 Apps from NERSC Report (Materials, Quantum Chem, High Energy Phys)… https://arxiv.org/abs/2509.09882v1

- How tight is the coupling (time sensitivity)?
- How much data is exchanged between QPU and CPU?
- What is the compute intensity of Quantum and Classical (HPC?)
- Can drive deployment, scheduling, parallelization decisions…